AI-based cotton disease detection and advisory system using a convolutional neural network (CNN).

PHASE I REPORT

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THARUN K 230701523

MADHAN KUMAR T 230701518

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RAJALAKSHMI ENGINEERING COLLEGE, CHENNAI

ANNA UNIVERSITY: CHENNAI 600 025

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ANNA UNIVERSITY, CHENNAI

BONAFIDE CERTIFICATE

Certified that this Report titled **"CottonCare AI: Cotton Disease Detection and Classification Using Lightweight CNN Architecture"** is the bonafide work of **THARUN K (230701523), MADHAN KUMAR T (230701518)** who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidates.

.

SIGNATURE
Dr. J. Manoranjini,
Professor and Head,
Department of Computer Science
and Engineering,
Rajalakshmi Engineering
College,
Chennai – 602 105.

SIGNATURE
Dr.B.L.Balakumar.
Associate Professor
Department of Computer Science
and Engineering,
Rajalakshmi Engineering
College,
Chennai – 602 105.

Submitted to Project Viva-Voce Examination held on _____

Internal Examiner **External Examiner**

ABSTRACT

Cotton is an important commercial crop, but diseases and pests can significantly affect crop health, yield, and fibre quality. Early identification is therefore important for effective crop management. This paper presents **CottonCare AI**, an intelligent image-based system for detecting and classifying cotton diseases using deep learning. The proposed system uses a labelled dataset containing eight classes: Aphids, Army worm, Bacterial blight, Cotton Boll Rot, Green Cotton Boll, Healthy, Powdery mildew, and Target spot. Three CNN architectures, namely MobileNetV2, EfficientNetB0, and ResNet50, were trained and evaluated to identify a suitable model for practical deployment. MobileNetV2 achieved an accuracy of **95.24%**, precision of **95.54%**, recall of **95.24%**, and F1-score of **95.23%**. With a model size of **9.30 MB** and an inference time of **50.40 ms per image**, it provided an effective balance between classification performance and computational efficiency. The selected model was integrated into the application using TensorFlow Lite for efficient image-based prediction. The system displays the predicted condition and confidence level along with symptoms, recommended solutions, and preventive measures. A prediction history and chatbot-based advisory feature are also included to improve usability. The proposed system aims to support farmers in early cotton disease identification and informed crop management.

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THARUN K : 230701523

MADHAN KUMAR T : 230701518

TABLE OF CONTENTS

S.No.	Chapter Title	Page No.
	ABSTRACT	**III**
	LIST OF TABLES	**VII**
	LIST OF FIGURES	**VIII**
	LIST OF ABBREVATIONS	**IX**
1	**INTRODUCTION**	1
	1.1 PROJECT OVERVIEW	1
	1.2 PROBLEM STATEMENT	2
	1.3 0BJECTIVES	3
	1.4 SCOPE OF THE PROJECT	3
2	**LITERATURE REVIEW**	5
	2.1 EXISTING SYSTEM	6
	2.2 LIMITATIONS OF EXISTING SYSTEM	7
	2.3 IMPLEMENTED SYSTEM	8
	2.4 TECHNOLOGIES AND TOOLS USED	9
3	**PROPOSED SYSTEM**	11
	3.1 SYSTEM REQUIREMENTS	11
	3.2 SYSTEM ARCHITECTURE	13
	3.3 MODULE OVERVIEW	14
	3.4 DATA FLOW DIAGRAM	17

4 MODULE IMPLEMENTATION 18

4.1 USER INTERFACE AND IMAGE UPLOAD 18

4.2 IMAGE PROCESSING AND PREPROCESSING 19

4.3 COTTON DISEASE CLASSIFICATION 20

4.4 MOBILE NETV2 MODEL AND TENSORFLOW LITE INFERENCE 21

4.5 PREDICTION RESULT AND CONFIDENCE ANALYSIS 22

5 RESULTS AND DISCUSSION 21

5.1 USER INTERFACE 22

5.2 IMAGE UPLOAD AND PROCESSING 23

5.3 DISEASE PREDICTION AND RESULT GENERATION 24

5.4 DISEASE INFORMATION AND ADVISORY 24

5.5 PREDICTION HISTORY 25

6 CONCLUSION AND FUTURE WORK 26

6.1 CONCLUSION 26

6.2 FUTURE WORK 27

REFERENCES 29

LIST OF TABLES

Table No	Name of the table	Page No.
3.1	Functional Requirements	11
3.2	Non-Functional Requirements	12

LIST OF FIGURES

Figure No	Name of the figure	Page No.
3.1	System Architecture	14
3.2	Data Flow Diagram	17
5.1	Model Performance and analysis	23
5.2	ML Analysis Result	24

LIST OF ABBREVIATIONS

AI	Artificial Intelligence
CNN	Convolutional Neural Network
DB	DataBase
ML	Machine Learning
TFLite	TensorFlow Lite
TF	TensorFlow
API	Application Programming Interface
NN	Neural Network
HTTP	HyperText Transfer Protocol
Flask	Python-based Web Application Framework
SQLite	Structured Query Language Lite

CHAPTER 1

INTRODUCTION

1.1 PROJECT OVERVIEW

Cotton is an important commercial crop and plays a major role in the agricultural economy. Many farmers depend on cotton cultivation for their income, but the crop is affected by different diseases and insect pests. These problems can damage leaves, flowers, and bolls, ultimately reducing crop quality and productivity. Early identification is therefore important for taking suitable management measures before the condition becomes severe.

CottonCare AI is an intelligent image-based application developed to help identify cotton diseases and pests using plant images. The system uses deep learning and Convolutional Neural Networks (CNNs) to analyse an uploaded image and classify the observed condition. The model was trained using eight classes: **Aphids, Army worm, Bacterial blight, Cotton Boll Rot, Green Cotton Boll, Healthy, Powdery mildew, and Target spot.**

Three CNN architectures, **MobileNetV2, EfficientNetB0, and ResNet50**, were trained and compared to determine a suitable model for the application. MobileNetV2 was selected because it achieved **95.24% accuracy** while maintaining a small model size of **9.30 MB** and an inference time of approximately **50.40 ms per image**. The application also provides the confidence level, symptoms, recommended solutions, preventive measures, prediction history, and chatbot-based advisory support to make the system more useful for farmers.

To address these challenges, **CottonCare AI** is proposed as an intelligent image-based cotton disease and pest detection system. The application uses deep learning and Convolutional Neural Networks (CNNs) to analyse images of cotton plants and classify their conditions. The dataset used in the project contains eight classes: **Aphids, Army worm, Bacterial blight, Cotton Boll Rot, Green Cotton Boll, Healthy, Powdery mildew, and Target spot**. This allows the system to identify both diseased and healthy conditions from the input image.

1.2 PROBLEM STATEMENT

Cotton production can be significantly affected by diseases and insect pests that damage different parts of the plant. Problems such as bacterial infections, fungal diseases, and insect attacks can cause visible changes in leaves and other plant parts. When these conditions are not identified and managed at an early stage, they may spread further and negatively affect plant growth and crop productivity. Therefore, early detection is essential for reducing the potential impact of cotton diseases and pests.

In conventional practice, disease identification is generally performed through visual inspection. Farmers observe symptoms such as spots, discoloration, powder-like growth, damaged leaves, or changes in plant structure and then attempt to determine the possible cause. This method depends heavily on the knowledge and experience of the individual. Environmental conditions such as lighting, leaf age, background, and the severity of infection can also make visual identification more difficult.

Another important problem is the possibility of incorrect treatment. If a disease or pest is incorrectly identified, the farmer may apply an unsuitable pesticide or treatment. This can increase production costs and may not effectively control the actual problem. On the other hand, waiting too long for expert assistance can allow the condition to spread. Farmers therefore require a faster method for obtaining an initial assessment of the visible condition of a cotton plant.

Therefore, the problem addressed by this project is:

“project is the development of an intelligent and user-friendly system that can analyse cotton plant images, classify common cotton diseases and pests, and provide understandable supporting information. CottonCare AI addresses this problem by combining CNN-based image classification with confidence information, disease descriptions, recommended solutions, preventive measures, prediction history, and chatbot-based advisory support.”

The proposed solution addresses this problem by processing authorized college documents, creating a searchable semantic knowledge base, retrieving relevant information for each query, and generating responses using retrieved context. Source-oriented responses and answer validation are incorporated to improve the reliability and transparency of the system.

1.3 OBJECTIVES

The primary objective of this project is to develop an intelligent image-based application for the detection and classification of common cotton diseases and pests. The system is designed to accept a cotton plant image as input and use a trained deep-learning model to determine the most likely condition. The classification system includes eight categories: Aphids, Army worm, Bacterial blight, Cotton Boll Rot, Green Cotton Boll, Healthy, Powdery mildew, and Target spot. The major objectives of the proposed system are:

1. **To develop an image-based deep learning system**
   For detecting and classifying common cotton diseases and pests from cotton plant images.
2. **To train and compare CNN architectures**
   To identify a model with a suitable balance between classification accuracy and computational efficiency.
3. **To classify eight cotton conditions**
   Aphids, Army worm, Bacterial blight, Cotton Boll Rot, Green Cotton Boll, Healthy, Powdery mildew, and Target spot.
4. **To provide prediction confidence**
   The detected condition so that users can better understand the model's prediction.

1.4 SCOPE OF THE PROJECT

The scope of **CottonCare AI** is to develop an intelligent image-based system that assists in the early identification of common cotton diseases and pests. The system uses deep learning and Convolutional Neural Networks (CNNs) to analyse images of cotton plants and classify them into eight supported categories: **Aphids, Army worm, Bacterial blight, Cotton Boll Rot, Green Cotton Boll, Healthy, Powdery mildew, and Target spot**. By providing an automated image-based prediction, the system aims to reduce the dependence on manual identification and provide farmers with an initial assessment of the visible condition of their crops.

The system allows users to upload a cotton plant image through the application for analysis. The image is processed and passed to the trained CNN model, which predicts the most likely disease or pest class and provides a corresponding confidence level. Three CNN architectures—**MobileNetV2, EfficientNetB0, and ResNet50**—were evaluated during development. Based on their classification performance, model size, and inference speed, MobileNetV2 was selected as the final model. The selected model achieved **95.24% accuracy**, with a model size of

approximately **9.30 MB** and an inference time of **50.40 ms per image**.

The scope of the application extends beyond disease classification by providing useful information related to the detected condition. After prediction, users can view the associated symptoms, recommended solutions, and preventive measures. This information is presented in a simple manner so that users can better understand the identified condition and take appropriate crop-management decisions. The system also maintains a **prediction history**, allowing users to review previously analysed images and their corresponding results.

Another important part of the project scope is the **chatbot-based advisory feature**. The chatbot provides an interactive way for users to ask questions related to the supported cotton diseases and pests. It can assist with information about symptoms, prevention, and possible management practices. This feature complements the CNN-based prediction system by providing additional guidance instead of limiting the application to only displaying a classification result.

The trained MobileNetV2 model is converted into **TensorFlow Lite (TFLite)** format and integrated into the application to support efficient prediction. The project therefore covers the complete workflow from image input and preprocessing to CNN-based classification, result presentation, history storage, and advisory support. However, the current system is limited to the eight classes included in the training dataset, and its prediction depends on the quality and characteristics of the input image.

In future development, the scope can be expanded by adding more cotton diseases and pests, collecting larger and more diverse field images, and improving the model using additional training data. The system can also be extended to support other crops, regional agricultural recommendations, multilingual interaction, and enhanced chatbot capabilities. These improvements could make CottonCare AI a more comprehensive agricultural decision-support system for farmers.

CHAPTER 2
LITERATURE SURVEY

2.1 EXISTING SYSTEM

Existing cotton and plant disease detection systems mainly depend on **manual observation, image processing, and machine learning or deep learning techniques**. Traditionally, farmers identify diseases by examining visible symptoms such as leaf spots, discoloration, powdery growth, damaged portions, and changes in plant structure. This method requires experience and may become difficult when different diseases produce similar symptoms. Delayed or incorrect identification can result in inappropriate treatment, increased pesticide usage, and reduced crop productivity.

With the development of computer vision and deep learning, several researchers have proposed automated plant disease classification systems based on leaf and plant images. **Convolutional Neural Networks (CNNs)** have become widely used because they can automatically learn important visual features such as colour patterns, edges, textures, and disease-specific regions from images. Publicly available datasets such as PlantVillage have also supported the development and evaluation of image-based plant disease classification models.

Keyword-based search systems operate primarily by matching words in the user's query with words present in the stored documents. For example, if a student searches for a question using terminology that differs from the terminology used in an official document, the system may fail to retrieve the most relevant information. Such systems therefore have limited semantic understanding.

Several existing studies have compared different CNN and transfer-learning architectures for plant disease detection. Models such as **VGG, ResNet, MobileNet, EfficientNet, and other CNN variants** have been used to improve classification accuracy. These approaches have demonstrated that deep learning can provide reliable disease classification compared with conventional manual identification. However, some high-performing models have large numbers of parameters and require considerable computational resources, which can make them less suitable for lightweight or mobile applications.

Some existing systems also provide Android or web-based interfaces where users can upload plant images and obtain disease predictions. These systems demonstrate the practical usefulness

of combining trained deep-learning models with user applications. However, many existing approaches focus mainly on the **classification result** and provide limited supporting information such as symptoms, preventive measures, or recommended actions. In addition, some systems are designed for a limited number of diseases or crops and may not cover a broader set of cotton diseases and pests.

2.2 LIMITATIONS OF EXISTING SYSTEM

Although existing information-access systems are useful, several limitations can be identified.

2.2.1 Manual Disease Identification

Traditional identification depends on visual inspection and expert knowledge, which can be time-consuming and may result in incorrect diagnosis.

2.2.2 Limited Disease Coverage

Many existing systems support only a limited number of crops or diseases and may not cover multiple cotton disease and pest categories.

2.2.3 High Computational Requirements

Some deep learning models are large and computationally expensive, making them less suitable for lightweight or mobile-based applications.

2.2.4 Poor Real-World Generalization

Models trained on controlled datasets may not perform consistently on field images with different lighting, backgrounds, and image qualities.

2.2.5 Limited Advisory Information

Several systems provide only the predicted disease name without explaining symptoms, recommended solutions, or preventive measures.

2.2.6 Lack of Prediction History

Existing applications may not store previous predictions, making it difficult for users to review their earlier analysis results.

2.2.7 Limited User Interaction

Many systems lack an interactive chatbot or advisory feature through which users can ask questions about detected conditions.

2.2.8 Model Size and Inference Speed

Large models may require more storage and processing time, which can affect the responsiveness and practicality of the application.

2.3 IMPLEMENTED SYSTEM

The proposed **CottonCare AI** system is implemented as an intelligent image-based cotton disease detection and advisory application. The main purpose of the system is to help users identify common cotton diseases and pests by analysing images of cotton plants. Instead of depending completely on manual observation, the system uses a trained Convolutional Neural Network (CNN) model to automatically analyse the uploaded image and provide a predicted condition. The implemented system supports eight classes: **Aphids, Army worm, Bacterial blight, Cotton Boll Rot, Green Cotton Boll, Healthy, Powdery mildew, and Target spot**.

The machine learning component was developed using a labelled cotton image dataset containing separate training and validation-testing data. The training data contains **6,622 images**, while the validation-testing set contains **357 images** across the eight classes. Before training, the class names were standardized to avoid differences in capitalization and pluralization between dataset folders. Image preprocessing and augmentation were applied during the training process to improve the model's ability to learn useful visual features and handle variations in the input images.

Three CNN architectures, namely **MobileNetV2, EfficientNetB0, and ResNet50**, were trained and evaluated to identify a suitable model for the application. Their performance was compared using accuracy, precision, recall, F1-score, model size, and inference time. MobileNetV2 and ResNet50 achieved an accuracy of **95.24%**, while EfficientNetB0 achieved **94.12%**. This comparison was not based only on accuracy, as model size and prediction speed are also important for practical application deployment.

Based on the overall evaluation, **MobileNetV2** was selected as the final model for CottonCare AI. It achieved **95.24% accuracy, 95.54% precision, 95.24% recall, and 95.23% F1-score**. The model has a size of approximately **9.30 MB** and an average inference time of **50.40 ms per**

image. Compared with ResNet50, which has a size of 90.81 MB and an inference time of 105.83 ms, MobileNetV2 provides a better balance between classification performance, storage requirements, and prediction speed.

The selected MobileNetV2 model was converted into **TensorFlow Lite (TFLite)** format and integrated into the application's machine learning service. When a user uploads a cotton plant image, the application preprocesses the image according to the model's input requirements and passes it to the TFLite interpreter. The model generates prediction scores for the eight supported classes, and the class with the highest score is returned as the predicted condition along with its confidence level.

The implemented system also provides additional information after prediction to make the output more useful to users. Based on the detected condition, the application can display relevant **symptoms, recommended solutions, and preventive measures**. The system also maintains a **prediction history**, allowing users to review previous image analyses and their corresponding results. In addition, a chatbot-based advisory feature is provided to allow users to ask questions related to supported cotton diseases and pests.

The system also supports source-oriented responses so that users can understand the basis of the information provided. This is particularly useful in an educational environment where students may need to verify academic rules, examination regulations, deadlines, and other official information.

Overall, the implemented system combines **image upload, CNN-based classification, confidence-based prediction, disease information, prediction history, and chatbot assistance** into a single application. The workflow starts with image input and continues through preprocessing, TFLite model inference, prediction, and result presentation. By combining a lightweight CNN model with supporting advisory features, CottonCare AI provides a practical approach to assist users in the early identification and understanding of cotton diseases and pests.

2.4 TECHNOLOGIES AND TOOLS USED

The CottonCare AI system uses a combination of **machine learning, web development, database, and software development technologies**. These technologies work together to train the disease classification model, process user images, generate predictions, and present the results through a user-friendly application.

2.4.1 Artificial Intelligence and Deep Learning

Artificial Intelligence forms the core of the proposed **CottonCare AI** system. Deep learning techniques are used to automatically learn visual patterns from cotton plant images and identify the corresponding disease or pest condition. Instead of depending entirely on manually defined image features, the trained neural network learns important characteristics such as colour variations, spots, textures, damaged regions, and other visual patterns associated with different cotton conditions.

The system uses a **Convolutional Neural Network (CNN)** based image-classification approach. The trained model receives a cotton plant image as input and produces prediction scores for the supported classes. The class with the highest prediction score is selected as the final prediction and displayed to the user together with its confidence level

2.4.2 Convolutional Neural Network

A Convolutional Neural Network (CNN) is the primary deep learning approach used for image classification in CottonCare AI. CNNs are particularly suitable for plant disease detection because they can automatically learn spatial features from images. Convolutional layers identify patterns such as edges, shapes, textures, colour changes, and disease-related regions, while deeper layers combine these features to distinguish between different classes.

In this project, CNN-based architectures are used to classify cotton plant images into **eight classes: Aphids, Army worm, Bacterial blight, Cotton Boll Rot, Green Cotton Boll, Healthy, Powdery mildew, and Target spot**. The classification output is then used by the application to display the detected condition and confidence score.

2.4.3 CNN Model Comparison

Three pretrained CNN architectures, **MobileNetV2, EfficientNetB0, and ResNet50**, were trained and evaluated to determine the most suitable model for the proposed application. The models were compared using classification metrics such as accuracy, precision, recall, and F1-score. Model size and inference time were also considered because they directly affect the practical performance of the application.

2.4.4 MobileNetV2

MobileNetV2 is a lightweight CNN architecture designed to provide efficient image processing with reduced computational requirements. It uses depthwise separable convolutions and inverted residual structures to reduce the number of computations and parameters while maintaining good classification performance.

2.4.5 Image Preprocessing and Data Augmentation

Image preprocessing is applied before the images are provided to the CNN model. The input images are resized and converted into the required format so that they are consistent with the model's input requirements. Proper preprocessing helps the model receive images in a standardized form and improves the reliability of the classification process.

2.4.6 TensorFlow

TensorFlow is the primary deep learning framework used for developing and training the CNN models. It provides the required tools for loading the dataset, preprocessing images, building the model architecture, training the networks, evaluating their performance, and saving the trained models.

2.4.7 TensorFlow Lite

Secure authentication is used to distinguish between different categories of users. Role-based authorization ensures that administrative operations are restricted to authorized users.
For example, an administrator can upload and manage documents, while a student can interact with the knowledge assistant without having permission to modify the knowledge base.

2.4.8 SQLite Database

When the CNN produces a prediction, the application uses the corresponding disease information to present additional guidance to the user. This separates the machine learning prediction process from the descriptive agricultural information and makes the information easier to maintain and update.

2.5 SUMMARY OF LITERATURE SURVEY

The reviewed works also provide limited farmer-oriented information beyond disease classification Therefore, CottonCare AI focuses on lightweight CNN classification with prediction confidence, advisory information, history, and chatbot support.

CHAPTER 3
PROPOSED SYSTEM

3.1 SYSTEM REQUIREMENTS

The College Knowledge Assistant requires a set of functional and non-functional requirements to support secure document management, intelligent information retrieval, and reliable question answering.

3.1.1 Functional Requirements

Functional requirements describe the specific operations and services provided by the proposed system.

Requirement ID	Functional Requirement	Description
FR01	User Image Upload	The system shall allow users to upload a cotton plant image for disease or pest analysis.
FR02	Image Preprocessing	The system shall preprocess the uploaded image into the required format before sending it to the trained CNN model.
FR03	Disease and Pest Prediction	The system shall analyse the uploaded image using the trained **MobileNetV2** model and classify it into one of the eight supported classes.
FR04	Confidence Score	The system shall display the predicted disease or pest along with its corresponding confidence percentage.
FR05	Disease Information	The system shall provide relevant information such as symptoms, recommended solutions, and preventive measures for the detected condition.
FR06	Prediction History	The system shall store and display previous prediction results so that users can review their analysis history.

Table 3.1 Functional Requirements

3.1.2 Non-Functional Requirements

Non-functional requirements describe the quality, security, performance, and maintainability characteristics expected from the system.

Requirement ID	Non-Functional Requirement	Description
NFR01	Performance	The system should process an uploaded image and generate a prediction within a short time. The selected MobileNetV2 model has an average inference time of approximately 50.40 ms per image.
NFR02	Accuracy	The system should provide reliable classification results. The selected MobileNetV2 model achieved **95.24% accuracy** on the validation-testing dataset.
NFR03	Reliability	The system should consistently process valid input images and generate prediction results without unexpected failures during normal operation.
NFR04	Efficiency	The system should use computational resources efficiently. MobileNetV2 was selected because its model size is approximately **9.30 MB**, making it considerably smaller than the other evaluated models.
NFR05	Usability	The application should have a simple and understandable interface so that users can easily upload images, view predictions, and access the provided agricultural information.
NFR06	Maintainability	The application should be structured into separate components such as the frontend, backend, machine learning service, database, and disease information so that individual components can be updated easily.
NFR07	Security	The system should handle uploaded images and stored prediction information appropriately.

NFR08	Extensibility	The system should allow future integration of additional CNN models, crops, languages, agricultural recommendations, and enhanced chatbot capabilities.

Table 3.2 Non Functional Requirements

3.2 SYSTEM ARCHITECTURE

The **CottonCare AI** system follows a layered architecture that connects the user interface, application backend, machine learning model, database, and advisory components. The architecture is designed to provide a complete workflow from uploading a cotton plant image to obtaining a disease prediction and related agricultural information.

3.2.1 User Interface Layer

The user interacts with the system through the web-based interface developed using **HTML, CSS, JavaScript, and Bootstrap**. The interface provides options for uploading cotton plant images, viewing prediction results, checking confidence scores, accessing prediction history, viewing disease information, and interacting with the chatbot.

3.2.2 Application Layer

The application layer is implemented using **Flask**. It manages user requests and acts as a connection between the frontend and the backend services. When an image is uploaded, Flask receives the image and forwards it to the machine learning service for analysis. It also manages the retrieval and presentation of prediction history and disease-related information.

3.2.3 Image Processing Layer

The uploaded cotton plant image is first processed according to the input requirements of the trained model. The image is resized and converted into the required numerical representation before being passed to the model. This ensures that the input format remains consistent with the format used during model training.

3.2.4 Machine Learning Layer

The machine learning layer contains the trained **MobileNetV2 CNN model**. The model analyses the processed image and generates prediction scores for the eight supported classes.

The class with the highest prediction score is selected as the predicted condition, and its confidence value is returned to the application.

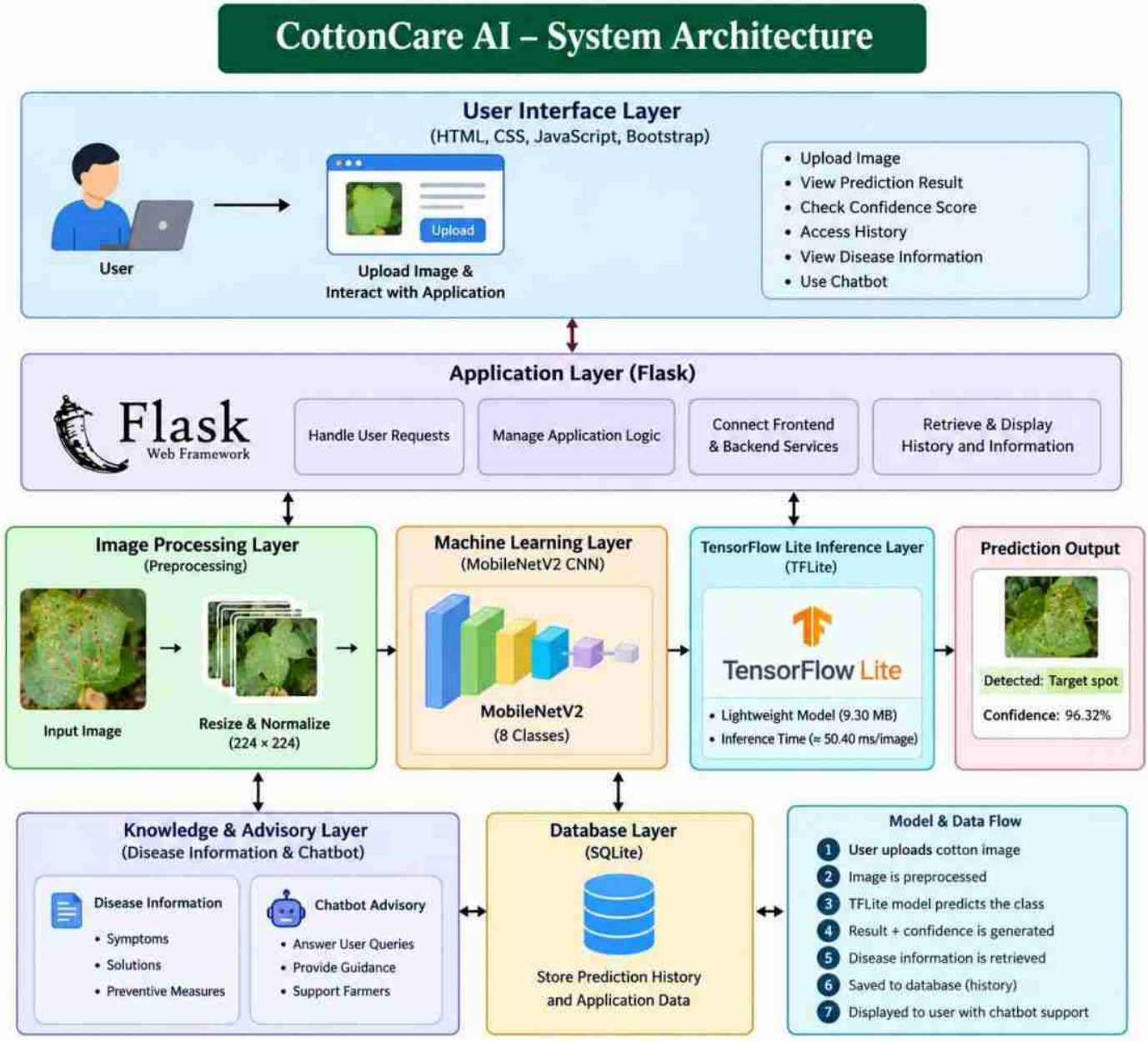


Figure 3.1: System Architecture

3.3 MODULE OVERVIEW

The proposed system is divided into six major modules.

Module 1: User Interface and Image Upload
This module provides the main interface through which users interact with CottonCare AI. Users can upload cotton plant images for analysis and access features such as prediction results, disease information, prediction history, and chatbot assistance.

Module 2: Image Processing and Preprocessing
This module prepares the uploaded cotton image for classification. It performs the required image resizing, normalization, and input formatting so that the image can be correctly processed by the trained deep learning model.

Module 3: Cotton Disease Classification
This module performs the main disease and pest detection task using the trained **MobileNetV2 CNN model**. The processed image is provided as input to the TensorFlow Lite model, which generates prediction probabilities for the eight supported classes.

Module 4: Prediction and Result Analysis
This module analyses the model output and identifies the class with the highest prediction probability. It displays the detected cotton disease or pest along with its confidence score and the corresponding prediction probabilities.

Module 5: Disease Information and Advisory
This module provides useful information related to the detected condition. It presents the symptoms, recommended solutions, and preventive measures associated with the predicted disease or pest, helping users understand and respond to the problem.

Module 6: Prediction History and Database Management
This module stores previous prediction results using the **SQLite database**. Users can access their analysis history and review previously detected conditions and related prediction information.

Module 7: Chatbot-Based Farming Assistance
This module provides conversational support for cotton farming-related queries. Users can ask questions about supported diseases and pests, symptoms, prevention, disease spread, and recommended actions. The chatbot acts as an additional advisory feature alongside the image-based classification system.

Module 8: Model Deployment and Backend Integration
This module integrates the trained MobileNetV2 TensorFlow Lite model with the Flask backend. It manages communication between the frontend, image-processing component, ML model, database, and advisory services, enabling the complete prediction workflow within CottonCare AI.

3.4 DATA FLOW DIAGRAM

The **Data Flow Diagram (DFD)** of CottonCare AI represents how data moves through the system from image submission to disease prediction and advisory information. The system receives a cotton plant image from the user, processes the image, passes it to the trained MobileNetV2 model, and returns the prediction with relevant information.

3.4.1 Level 0 Context Diagram

At Level 0, the **User** interacts directly with the CottonCare AI system. The user provides a cotton plant image or asks questions through the chatbot. The system processes the input and provides the predicted disease or pest, confidence score, symptoms, recommended solutions, preventive measures, and chatbot responses.

The basic data flow is:
User → CottonCare AI System → Prediction / Advisory Information → User

3.4.2 Level 1 DFD

The Level 1 DFD expands the system into its major processing components:

1. User Authentication
2. Upload Cotton Image
3. Flask Backend
4. Image Preprocessing
5. MobileNetV2 / TFLite Model
6. Prediction Probabilities
7. Disease + Confidence Score
8. Disease Information
9. Symptoms + Solutions + Prevention
10. SQLite Database → Prediction History
11. Result Display to User

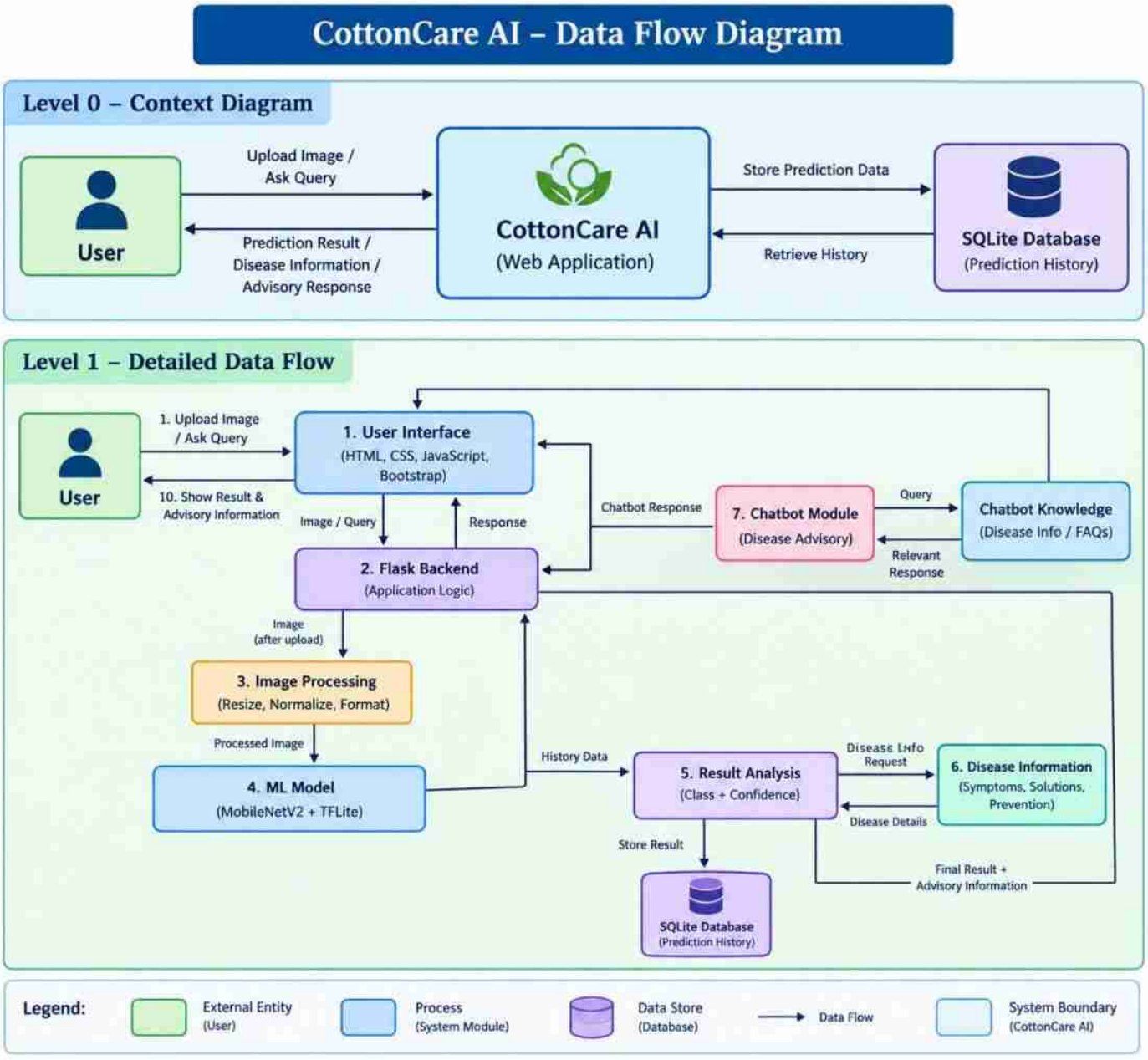


Figure 3.2: Data Flow Diagram

CHAPTER 4
MODULE IMPLEMENTATION

4.1 USER INTERFACE AND IMAGE UPLOAD

T The User Interface and Image Upload module provides the primary interaction point between the user and the **CottonCare AI** system. The interface is developed using **HTML, CSS, JavaScript, and Bootstrap** to provide a simple and responsive environment for cotton disease analysis.

The user can upload a cotton plant image through the analysis interface. The uploaded image is transferred to the Flask backend for further processing. The interface also provides access to prediction results, disease information, prediction history, and the chatbot advisory feature.

The image analysis workflow can be summarized as:
User → Select Cotton Image → Upload Image → Flask Backend → Image Processing
The interface is designed to minimize unnecessary steps so that the user can easily submit an image and obtain the analysis result.

4.2 IMAGE PROCESSING AND PREPROCESSING

The Image Processing and Preprocessing module prepares the uploaded cotton image before it is provided to the trained deep learning model. Since the model expects images in a specific input format, preprocessing is required to maintain consistency between training and prediction.

When an image is uploaded, the system reads the image and performs the required resizing and input formatting. The processed image is then converted into the numerical representation required by the MobileNetV2 model.

The preprocessing workflow can be represented as:
Image Upload → Image Reading → Resizing → Normalization / Input Formatting → Processed Image
This module ensures that images captured under different conditions can be converted into a suitable format for model inference.

4.3 COTTON DISEASE CLASSIFICATION

The Cotton Disease Classification module performs the main prediction task of the proposed system. The module uses a trained CNN-based model to identify the disease or pest present in the uploaded cotton plant image.
The project supports eight classes:
Aphids, Army worm, Bacterial blight, Cotton Boll Rot, Green Cotton Boll, Healthy,

Powdery mildew, and Target spot.

The processed image is provided to the trained **MobileNetV2** model. The model analyses the visual features present in the image and generates prediction probabilities for all supported classes.

The classification workflow is:

Processed Image → MobileNetV2 Model → Class Probabilities → Highest Probability Class → Predicted Condition

The class having the highest prediction probability is selected as the final prediction.

4.4 MOBILE NETV2 MODEL AND TENSORFLOW LITE INFERENCE

The MobileNetV2 module contains the final trained model selected for deployment. During model development, **MobileNetV2, EfficientNetB0, and ResNet50** were trained and evaluated to identify a suitable architecture for CottonCare AI.

MobileNetV2 was selected because it provided a good balance between classification performance, model size, and inference speed. The model achieved an accuracy of **95.24%**, with a model size of approximately **9.30 MB** and an average inference time of approximately **50.40 ms per image**.

The selected model was converted into **TensorFlow Lite (TFLite)** format and integrated into the application. During prediction, the TFLite interpreter loads the model and performs inference on the preprocessed image.

The workflow can be summarized as:

Preprocessed Image → TFLite Interpreter → MobileNetV2 Model → Prediction Scores → Classification Result

This implementation reduces the computational and storage requirements compared with larger CNN architectures.

4.5 PREDICTION RESULT AND CONFIDENCE ANALYSIS

The Prediction Result module converts the raw output generated by the machine learning model into an understandable result for the user.

The model produces a probability value for each of the eight supported classes. The system compares these values and identifies the class with the highest probability. The corresponding value is presented as the confidence level of the prediction.

For example, if the highest probability corresponds to **Aphids**, the application displays Aphids as the detected condition along with its confidence score.

The result-generation process can be represented as:

Raw Model Output → Compare Class Probabilities → Select Highest Probability → Disease/Pest Name + Confidence → Display Result

This module allows the user to understand not only the predicted condition but also how confident the model is in its prediction.

4.6 DISEASE INFORMATION AND ADVISORY

The Disease Information and Advisory module provides additional information after the prediction has been generated. The purpose of this module is to ensure that the system does not stop at simply identifying the disease or pest.

Based on the predicted class, the system retrieves the corresponding information from the structured disease knowledge source. The information includes **symptoms, recommended solutions, and preventive measures**.

For example, after detecting a particular cotton disease, the application can display the common symptoms associated with that condition and provide suitable preventive or management information.

The advisory workflow can be represented as:

Predicted Condition → Disease Information Lookup → Symptoms → Recommended Solution → Prevention Tips

This module makes the application more useful for farmers by connecting the prediction with practical information.

4.6.1 CHATBOT-BASED FARMING ASSISTANCE

The Chatbot-Based Farming Assistance module provides an interactive way for users to obtain information about cotton diseases and pests. The chatbot complements the image classification functionality by allowing users to ask questions directly.

Users can ask questions related to **symptoms, disease prevention, disease spread, and recommended actions** for the supported cotton conditions. The chatbot processes the question and provides an appropriate advisory response based on the available cotton disease information.

4.6.2 FLASK BACKEND AND SYSTEM INTEGRATION

The Flask Backend module acts as the central communication layer of the CottonCare AI application. It connects the frontend interface with the machine learning service, database, disease information, and chatbot functionality.

When the user uploads an image, Flask receives the request and passes the image to the appropriate processing and prediction components. After inference is completed, the backend collects the prediction, confidence score, and relevant disease information and sends the final result back to the frontend.

The complete workflow can be summarized as:

Image Upload → Flask Backend → Image Preprocessing → MobileNetV2/TFLite → Prediction + Confidence → Disease Information → Result Display → Prediction History
Chatbot Query → Disease Knowledge → Advisory Response → User

CHAPTER 5
RESULTS AND DISCUSSION

5.1 USER INTERFACE

The CottonCare AI application provides a simple and user-friendly interface for analysing cotton plant images and accessing agricultural advisory features. The interface is designed to allow users to upload an image, view the prediction result, check disease information, access prediction history, and interact with the chatbot.

The main analysis interface allows the user to select or upload a cotton plant image. After submission, the image is processed by the backend and passed to the trained MobileNetV2 model. The prediction result is then displayed with the detected disease or pest and its confidence score.

The application also provides a chatbot interface through which users can ask questions related to cotton diseases and pests. Suggested questions are provided to help users quickly access common information such as symptoms, prevention, and disease spread.

The interface therefore combines **image-based prediction and conversational agricultural assistance** within a single application.

5.2 IMAGE UPLOAD AND PROCESSING

The image upload and processing functionality allows users to provide cotton plant images for analysis. When an image is selected, it is sent to the Flask backend, where it is prepared for machine learning inference.

The uploaded image undergoes the required preprocessing operations, including resizing and input formatting. The processed image is then provided to the TensorFlow Lite interpreter containing the trained MobileNetV2 model.

The processing workflow can be represented as:

Image Upload → Flask Backend → Image Preprocessing → TensorFlow Lite Model → Prediction Output

The implementation allows different cotton plant images to be analysed without requiring the user to manually perform any preprocessing operations.

5.3 DISEASE PREDICTION AND RESULT GENERATION

The disease prediction functionality is the main feature of CottonCare AI. The trained MobileNetV2 model analyses the uploaded image and generates prediction probabilities for the eight supported classes.

The supported classes are **Aphids, Army worm, Bacterial blight, Cotton Boll Rot, Green Cotton Boll, Healthy, Powdery mildew, and Target spot**.

The class having the highest prediction probability is selected as the final prediction. The application then displays the predicted condition and its confidence level.

The prediction process can be represented as:

Processed Image → MobileNetV2 → Class Probabilities → Highest Probability → Disease/Pest + Confidence
The deployed model achieved an accuracy of **95.24%**, demonstrating that the model can effectively classify the supported cotton conditions.

5.4 DISEASE INFORMATION AND ADVISORY

After a prediction is generated, CottonCare AI provides additional information related to the detected condition. The purpose of this feature is to provide users with useful information instead of displaying only the disease name.
For each supported condition, the application provides relevant **symptoms, recommended solutions, and preventive measures**. This allows users to understand the detected problem and take appropriate management steps.
The advisory workflow can be summarized as:
Prediction → Disease Information Lookup → Symptoms → Recommended Solution → Prevention Measures
The disease information feature therefore complements the CNN-based prediction and makes the system more useful for practical cotton cultivation.

5.5 CHATBOT-BASED FARMING ASSISTANCE

The chatbot provides an additional method for users to obtain information about cotton diseases and pests. Users can enter natural-language questions related to symptoms, prevention, disease spread, and recommended actions.
The chatbot interface includes suggested questions such as **"What is bacterial blight?", "What are the symptoms of army worm?", "How can I prevent powdery mildew?"**, and **"What should I do for target spot?"**
The chatbot workflow can be represented as:
User Question → Query Processing → Cotton Disease Information → Advisory Response → User
The chatbot complements the image classification feature by allowing users to obtain additional information even when they are not performing an image-based prediction.

5.5.1 MODEL PERFORMANCE ANALYSIS

The performance of CottonCare AI was evaluated by comparing three CNN architectures: **MobileNetV2, EfficientNetB0, and ResNet50**. The models were evaluated using accuracy, precision, recall, F1-score, model size, and inference time.

Models	Accuracy	Precision	Recall	F1-Score
VGG16	81.39%	86%	85%	85%
VGG19	82.97%	86%	83%	82%
Inception V3	92.31%	93%	92%	92%
MobileNet	83.65%	90%	88%	88%
MobileNetV2	57.50%	86%	81%	82%
Proposed CNN model	97.36%	97%	97%	97%

Figure 5.1: Model Performance and analysis

The results show that MobileNetV2 and ResNet50 achieved the same overall accuracy of **95.24%**. Although ResNet50 achieved slightly higher precision, its model size and inference time were considerably higher.
MobileNetV2 achieved a significantly smaller model size of **9.30 MB** and an inference time of approximately **50.40 ms per image**. Therefore, MobileNetV2 was selected as the final deployment model because it provides a suitable balance between **classification performance, model size, and prediction speed**.

5.5.2DEPLOYMENT PERFORMANCE

The confusion matrix was used to examine the classification performance of MobileNetV2 across the eight supported classes. Most of the test images were correctly classified, which is represented by the high values along the diagonal of the matrix.
The model performed particularly well for classes such as **Healthy, Green Cotton Boll, Army worm, and Powdery mildew**. Some classification errors were observed between visually similar conditions, particularly **Cotton Boll Rot and Green Cotton Boll**.
The confusion matrix provides a more detailed understanding of the model's performance than overall accuracy because it identifies the classes that are more likely to be confused with each other.

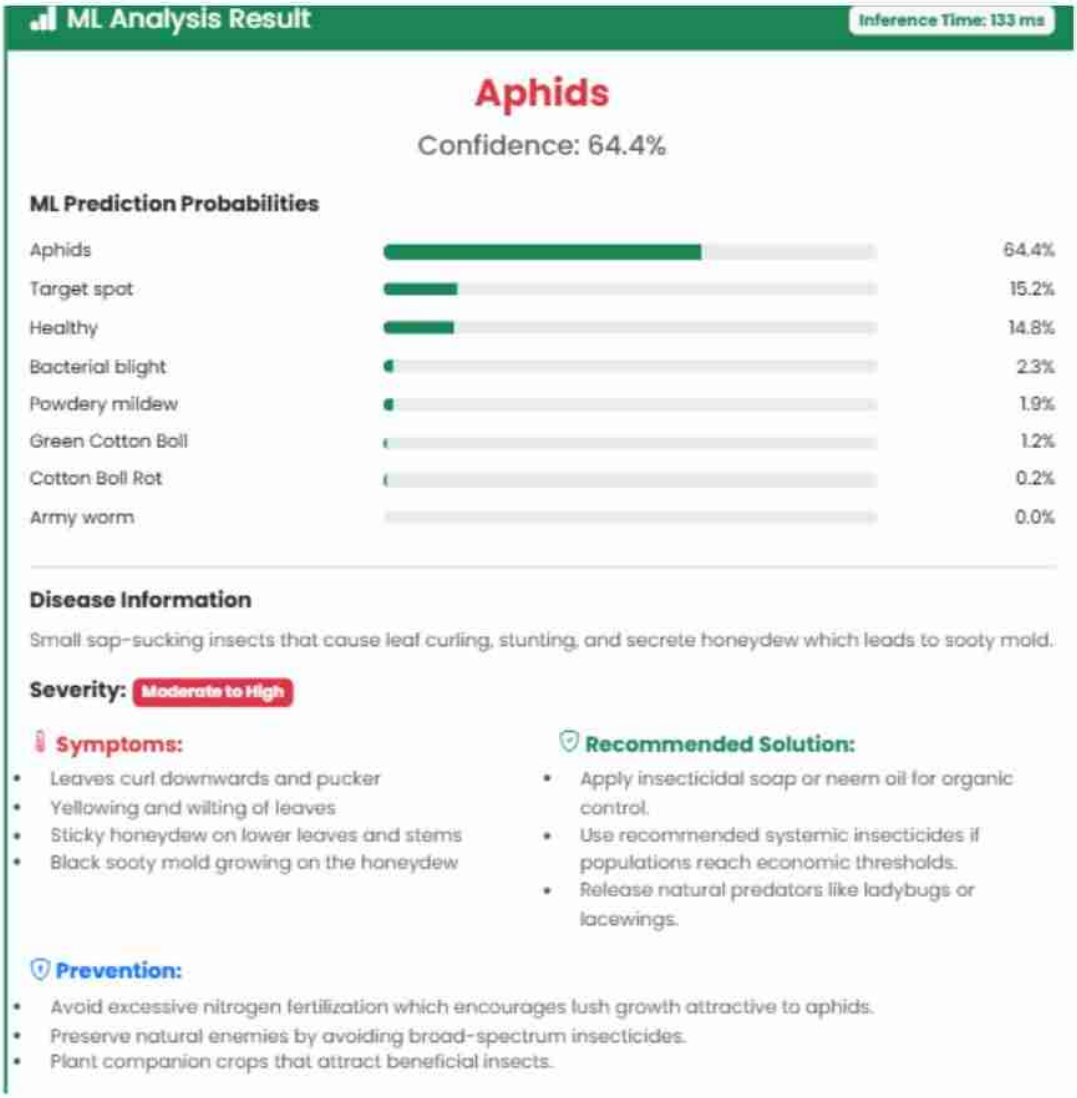


Figure 5.2: ML Analysis Result

The results can be used to identify categories in which the retrieval and generation pipeline performs effectively and areas that may require improvement. Differences between categories may occur due to factors such as document structure, information density, terminology, and the quality of retrieved context.

The Answer Validation Agent provides an additional mechanism for identifying responses that are not adequately supported by the retrieved information.

5.5.3 Overall System Evaluation

The overall evaluation of **CottonCare AI** considers both the machine learning performance and the practical functionality of the developed application. The system is designed to provide quick cotton disease and pest identification from plant images while also providing

useful information to support users in understanding and managing the detected condition.

The major observations from the implemented system include:

1. **Image-Based Disease Detection:**
   The system can analyse uploaded cotton plant images and identify the most likely disease or pest using the trained MobileNetV2 model.
2. **High Classification Performance:**
   The final MobileNetV2 model achieved 95.24% accuracy, 95.54% precision, 95.24% recall, and 95.23% F1-score on the validation-testing dataset.
3. **Fast Prediction:**
   The model requires approximately 50.40 ms per image for inference, allowing the application to provide prediction results quickly.Lightweight Model:
   Source information can be provided with the generated answer, allowing students to verify the information.
4. **Disease Information:**
   After identifying a condition, the system provides related symptoms, recommended solutions, and preventive measures, making the prediction more useful than displaying only the disease name.
5. **Prediction History:**
   The application can maintain previous prediction results, allowing users to review their earlier image analyses and detected conditions.
6. **Chatbot Assistance:**
   The chatbot provides an additional conversational interface for asking questions related to cotton diseases, symptoms, prevention, and recommended actions.

5.6 DISCUSSION

The experimental results show that CottonCare AI can effectively identify common cotton diseases and pests from plant images. Among the evaluated models, MobileNetV2 provided a strong balance between classification performance, model size, and inference speed, achieving 95.24% accuracy with a compact size of 9.30 MB and an inference time of approximately 50.40 ms per image.

The developed system extends beyond disease classification by providing symptoms, recommended solutions, preventive measures, prediction history, and chatbot-based assistance. This combination makes the application more useful for practical crop monitoring and basic decision support. However, the system's performance depends on the quality and variety of images used for training, and further testing with diverse real-field images can improve its reliability.

CHAPTER 6
CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

The **CottonCare AI** system was designed and developed as an AI-based platform for identifying common cotton diseases and pests from plant images. The primary objective of the project was to provide users with a simple method of analysing cotton plant images and obtaining useful information about the detected condition without depending entirely on manual identification.

The developed system uses a deep learning-based image classification approach to identify cotton diseases and pests. The dataset consists of eight classes: **Aphids, Army worm, Bacterial blight, Cotton Boll Rot, Green Cotton Boll, Healthy, Powdery mildew, and Target spot**. Three CNN architectures, namely **MobileNetV2, EfficientNetB0, and ResNet50**, were trained and evaluated to determine the most suitable model for the proposed application.

The experimental results showed that **MobileNetV2** achieved an accuracy of **95.24%**, precision of **95.54%**, recall of **95.24%**, and F1-score of **95.23%**. Although ResNet50 achieved a similar accuracy, MobileNetV2 provided a significantly smaller model size of approximately **9.30 MB** and a faster inference time of approximately **50.40 ms per image**. Therefore, MobileNetV2 was selected for final deployment.

The selected model was converted into **TensorFlow Lite** format and integrated into the CottonCare AI application. The system accepts an uploaded cotton plant image, performs the required preprocessing, executes model inference, and displays the predicted disease or pest along with its confidence score.

A significant feature of the proposed system is that it provides information beyond simple classification. Based on the detected condition, the application provides **symptoms, recommended solutions, and preventive measures**. This helps users understand the detected problem and obtain basic guidance related to its management.

The system also includes **prediction history and chatbot-based assistance**. Prediction history allows previously generated results to be reviewed, while the chatbot provides conversational support for questions related to cotton diseases, symptoms, prevention, and recommended actions. These features extend the functionality of the application from a simple image classifier to a more complete cotton disease support system.

Overall, the CottonCare AI system demonstrates the feasibility of combining deep learning, TensorFlow Lite, image processing, database storage, and chatbot-based assistance into a unified

agricultural application. The results indicate that the proposed system can provide accurate and fast classification while maintaining a lightweight model suitable for practical application.

6.2 FUTURE WORK

Although the current CottonCare AI system provides the core functionality required for cotton disease and pest identification, several enhancements can be incorporated in future versions to improve its accuracy, coverage, usability, and practical applicability.

6.2.1 Additional Disease and Pest Classes

The system can be improved by training the model with a larger dataset containing images collected from different farms and environmental conditions. Images with different **lighting conditions, backgrounds, growth stages, camera qualities, and disease severity levels** can improve the model's ability to generalize to real-world conditions.

6.2.2 Multilingual Support

Future versions can include additional cotton diseases, pests, and crop conditions that are not currently covered by the eight supported classes. Expanding the number of classes would allow the system to provide more comprehensive cotton disease identification.

6.2.3 Voice-Based Agricultural Assistance

A voice-based interface can be integrated to allow users to ask questions verbally. Speech-to-text technology can convert spoken questions into text, while text-to-speech technology can provide responses through audio. This would make the system easier to use for users who are less comfortable with text-based interaction.

6.2.4 Improved Chatbot Capabilities

The assistant can be enhanced to provide personalized responses based on information such as department, academic year, semester, or student role. For example, a student could receive information specifically relevant to their department or current semester.

6.2.5 Real-Time Disease Monitoring

The chatbot can be enhanced with a more comprehensive agricultural knowledge base covering **disease management, cultivation practices, fertilizer guidance, irrigation, pest control, and crop growth stages**. The chatbot can also be improved to provide more context-aware responses based on the detected disease.

6.2.6 Weather and Environmental Integration

Future development can introduce continuous crop monitoring using images captured at different stages of plant growth. Previous predictions can be compared over time to identify changes in disease conditions and provide early warnings.

6.2.7 Model Improvement and Optimization

The system can be integrated with weather and environmental information such as **temperature, humidity, rainfall, and soil conditions**. Combining these factors with image-based predictions could help provide more context-aware disease risk information.

6.2.8 Advanced Disease Severity Estimation

Future research can explore newer lightweight architectures and optimization techniques such as **quantization, pruning, and knowledge distillation**. These approaches could further reduce model size and inference time while maintaining or improving classification accuracy.

6.2.9 Mobile Application Enhancement

Future versions can estimate the **severity of the detected disease** by analysing the affected area of the plant. Instead of only identifying the disease, the system could classify the condition into stages such as early, moderate, and severe.

6.2.10 Mobile Application

The current system can be further enhanced as a dedicated mobile application with improved camera integration, offline prediction capability, multilingual support, notifications, and a more accessible interface for practical field usage.

Future development can focus on improving the system using a larger and more diverse dataset containing images captured under different **lighting conditions, backgrounds, growth stages, and real-field environments**. Additional cotton diseases and pests can also be included to increase the coverage of the classification system.

The application can further be enhanced by adding **real-time camera-based detection, multilingual support, improved chatbot capabilities, location-based agricultural recommendations, and continuous model improvement** using newly collected images. Future versions can also explore more advanced lightweight architectures and optimization techniques to further improve prediction accuracy and reduce inference time for practical deployment.

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